

Remove Deprecated `shared_ptr` Atomic Access APIs From C++26

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1 Abstract

Annex D of the C++ Standard, deprecated features, maintains an easily misused API for atomic access to `shared_ptr` objects. This paper proposes removing that API from the C++ Standard Library.

2 Revision history

R2: 2023 September (midterm)

— Removed revision history’s redundant subsection numbering

- Applied numerous editorial corrections
- Add a new option to resolve header issue with new function overloads
- Wording updates
 - Rebased onto latest working draft, [\[N4958\]](#)
 - Updated stable label cross-reference to C++23

R1: 2023 August (midterm)

- Recorded review feedback from SG1, recommending removal
- Moved from SG1 to LEWG queue
- Fixed grammar and presentation of some rationale, no functional change
- Revised rationale in Annex C
- Validated wording against latest Standard working draft, N4950

R0: 2023 May (pre-Varna)

Original version of this document, extracted from the C++23 proposal [\[P2139R2\]](#).

Key changes since that earlier paper:

- Rebased wording onto working draft N4944
- Added examples of how to update deprecated code
- Considered proposals to minimize impact on header usage
- Added Annex C wording

3 Introduction

At the start of the C++23 cycle, [\[P2139R2\]](#) tried to review each deprecated feature of C++ to see which we would benefit from actively removing and which might now be better undeprecated. Consolidating all this analysis into one place was intended to ease the (L)EWG review process but in return gave the author so much feedback that the next revision of the paper was not completed.

For the C++26 cycle, a much shorter paper, [\[P2863R1\]](#), will track the overall analysis, but for features that the author wants to actively progress, a distinct paper will decouple progress from the larger paper so that the delays on a single feature do not hold up progress on all.

This paper takes up the deprecated C-style API for race-free access to `shared_ptr` objects, D.23 [\[depr.util.smartptr.shared.atomic\]](#).

4 History

This removal was originally suggested for C++23 as part of [\[P2139R2\]](#), and at the LEWG telecon of 2020/07/13 was deferred (without technical discussion) to SG1 for its initial review, after which the removal discussion would have come back to LEWG. That initial review did not occur, so this paper has been produced for C++26 to enable easier tracking of each deprecated topic.

4.1 Origin

The free-function API for atomic access to `shared_ptr` was introduced with C++11, which introduced both the concurrency-aware memory model (including atomics) and `shared_ptr`.

4.2 Deprecation

The API was first deprecated by C++20, along with the introduction of its type-safe replacement, `atomic<shared_ptr<T>>`.

5 Proposal

It is now time to complete the cycle and remove the original fragile facility.

The legacy C-style atomic API for manipulating shared pointers, provided since C++11, is subtle and frequently misunderstood: a `shared_ptr` object that is to be used with the atomic API can never be used directly and (other than construction and destruction) may be manipulated **only** through the atomic API. Its failure mode on misuse (any direct use of that `shared_ptr` object before, after, or concurrent with the first use of the atomic access API) is silently undefined behavior, typically producing a data race.

C++20 provides `atomic<shared_ptr<T>>`, a type-safe alternative that encapsulates its `shared_ptr` object, safely providing a complete replacement for the original functionality'. Additionally, C++20 also provides support for `atomic<weak_ptr<T>>`.

5.1 Impact of Removal

The Standard contains no other overloads for the C style atomics interface taking pointers to `T` rather than pointers to `atomic<T>`, so all existing usage should be easily diagnosed by recompiling (if not already diagnosed by a deprecation warning today). The fix for old code should be as simple as replacing `shared_ptr<T>` with `atomic<shared_ptr<T>>` in the affected places. The existing C-style atomic interface should then pick up support for the `atomic<shared_ptr<T>>` type.

For example, consider migrating this legal (but deprecated) program from the original C++11 API to the type safe C++20 form:

Deprecated	Supported
<pre>#include <memory> std::shared_ptr<int> x; int main() { std::shared_ptr<int> y = std::atomic_load(&x); y.reset(new int(42)); std::atomic_store(&x, y); }</pre>	<pre>#include <memory> #include <atomic> std::atomic<std::shared_ptr<int>> x; int main() { std::shared_ptr<int> y = std::atomic_load(&x); y.reset(new int(42)); std::atomic_store(&x, y); }</pre>

Observe that only the global variable is changed by wrapping it in a `std::atomic`. No further changes to the code are necessary since the existing overloads for the C-style API expect `std::atomic<T>` pointers in the same argument positions and those calls provide the correct behavior.

Note we must also `#include` the `<atomic>` header as the (never deprecated) C-style API for atomics is defined in that header, once the deprecated overloads for `shared_ptr` have been removed from `<memory>`.

Alternatively, the user may prefer to further refactor the code to use the `std::atomic` member functions directly:

Deprecated	Refactored
<pre>#include <memory> std::shared_ptr<int> x; int main() { std::shared_ptr<int> y = std::atomic_load(&x); y.reset(new int(42)); std::atomic_store(&x, y); }</pre>	<pre>#include <memory> std::atomic<std::shared_ptr<int>> x; int main() { std::shared_ptr<int> y = x.load(); y.reset(new int(42)); x.store(y); }</pre>

While this refactored example contains more changes, one might argue that the example shows more idiomatic C++. Also, the header dependencies remain the same as the original code, as the full specification for `atomic<shared_ptr<T>>` is in the `<memory>` header needed for the original use of `shared_ptr`.

5.2 Addressing the Header Dependency

One concern when migrating to type-safe use of `atomic<shared_ptr<T>>` is that the overloaded functions for atomic types are declared only in the `<atomic>` header. The supposedly obvious solution would be to add the relevant atomic overloads that correspond to the old `<shared_ptr>` API. Wording for this solution is provided below, but what are the precedents and concerns? The following directions are considered, in order of increasing visibility of declarations through the `<memory>` header. Note that none of these concerns apply when importing the standard library modules.

5.2.1 Leave to user

The simplest option is to take no action in the standard specification, and leave the workaround to end users including additional headers as required.

If we review QoI of existing implementations, we find that MSVC already implicitly provides the API from just including `<memory>`; the GCC libstdc++ library strictly requires users to include `<atomic>` for themselves; the LLVM libc++ library does not yet implement this C++20 library.

We recommend against this direction. While the author has an aesthetic distaste for the way the container API has leaked across headers, in practice the wording below seems like a practical solution to simplify the process of updating code when the deprecated API is removed.

5.2.2 Add new `atomic<shared_ptr>` free-functions to `<memory>`

The free function interface for `atomic<shared_ptr>` is a subset of the free function interface for `atomic` objects in general, due to the lack of support for `volatile` overloads in the `shared_ptr` interface.

Rather than import the generic overloads from the `<atomic>` header, we could add new function template overloads to the `<memory>` header along side the declaration of `atomic<shared_ptr<T>>`, that specifically take `atomic<shared_ptr<T>>` parameters rather than `atomic<T>`. These overloads would be defined to have the same behavior as the corresponding overloads for `atomic<T>`.

5.2.3 Add minimal atomic free-functions to `<memory>`

The subset of atomic overloads could be added to the `<memory>` header along side the declaration of `atomic<shared_ptr<T>>`, while the specification remains untouched in the atomics part of the library.

The obvious precedent for declaring a set of functions in multiple headers is the set of container overloads in the `<iterator>` header, such as `begin`, `end`, and `data`. The same overloads are present in each container header

so that clients of that container can easily use these functions; however, the specification for these functions remains in the iterators part of the Standard.

5.2.4 Add all atomic free-functions to `<memory>`

The chief concern with adding just the minimal set of overloads is that, while containing all the overloads necessary to support the `shared_ptr` API, that set is just a subset of the complete set of overloaded declarations in the `<atomic>` header, notably missing all pointer-to-volatile overloads, and those functions that would be ill-formed for `shared_ptr`, so constrained to not exist for such instantiations.

If we are worried about that partial overload set, another option would be to add all the free-function interface of the `<atomic>` header to `<memory>`. The author believes that to be an excessive creep of unnecessary functionality into another header.

Given that the primary template `atomic<T>` cannot be instantiated for types other than instantiations of `shared_ptr` and `weak_ptr` without also including the `<atomic>` header, it abusing this partial overload set seems tricky in practice.

5.2.5 Include `<atomic>` from `<memory>`

A simpler and more practical approach might be to simply mandate that the `<memory>` header directly includes `<atomic>`, just as it already includes `<compare>`. While this solution seems to be a bigger leak of excessive functionality through an unrelated header, in practice the implementation of `shared_ptr` requires the use of atomic integers to handle the strong and weak reference counts. Nevertheless, this approach does seem to be a more impactful change than necessary with potential to impact compile times.

6 Review

6.1 SG1 Review : Varna 2023

SG1 reviewed this paper at the 2023 Varna meeting, and saw no concerns.

Poll: Remove deprecated `shared_ptr` atomic access APIs from C++25, with any of the library options listed in P2689.

SF	F	N	A	SA
2	4	1	1	0

The one vote against was a principled concern about any removal of deprecated features being a breaking change — no special concerns about this specific paper.

Forward LEWG to make the final design decisions on how best to handle the header compatibility issue.

7 Wording

Make the following changes to the C++ Working Draft. All wording is relative to [N4958], the latest draft at the time of writing. This wording takes the [minimal overloads](#) approach to adding declarations to the `<memory>` header.

7.1 Add declarations to the synopsis for the header `<memory>`

7.1.1 20.2.2 [\[memory.syn\]](#) Header `<memory>` synopsis

```
namespace std {  
    // ...  
  
    // 33.5.8.7[util.smartptr.atomic], atomic smart pointers
```

```

template<class T> struct atomic; // freestanding
template<class T> struct atomic<shared_ptr<T>>;
template<class T> struct atomic<weak_ptr<T>>;

// 33.5.9[atomics.nonmembers], atomic non-member functions
template <class T>
    bool atomic_is_lock_free(const atomic<T>*) noexcept; // freestanding

template <class T>
    T atomic_load(const atomic<T>*) noexcept; // freestanding
template <class T>
    T atomic_load_explicit(const atomic<T>*, memory_order) noexcept; // freestanding

template <class T>
    void atomic_store(atomic<T>*, typename atomic<T>::value_type) noexcept; // freestanding
template <class T>
    void atomic_store_explicit(atomic<T>*, // freestanding
                               typename atomic<T>::value_type,
                               memory_order) noexcept;

template <class T>
    T atomic_exchange(atomic<T>*, typename atomic<T>::value_type) noexcept; // freestanding
template <class T>
    T atomic_exchange_explicit(atomic<T>*, // freestanding
                               typename atomic<T>::value_type,
                               memory_order) noexcept;

template <class T>
    bool atomic_compare_exchange_weak(atomic<T>*, // freestanding
                                       typename atomic<T>::value_type*,
                                       typename atomic<T>::value_type) noexcept;

template <class T>
    bool atomic_compare_exchange_strong(atomic<T>*, // freestanding
                                       typename atomic<T>::value_type*,
                                       typename atomic<T>::value_type) noexcept;

template <class T>
    bool atomic_compare_exchange_weak_explicit(atomic<T>*, // freestanding
                                                typename atomic<T>::value_type*,
                                                typename atomic<T>::value_type,
                                                memory_order, memory_order) noexcept;

template <class T>
    bool atomic_compare_exchange_strong_explicit(atomic<T>*, // freestanding
                                                  typename atomic<T>::value_type*,
                                                  typename atomic<T>::value_type,
                                                  memory_order, memory_order) noexcept;

// 20.3.4.1[out_ptr.t], class template out_ptr_t
template<class Smart, class Pointer, class... Args>
    class out_ptr_t;

// ...
}

```

7.2 Update Annex C

Annex C (informative) Compatibility [diff]

C.1.X Annex D: compatibility features [diff.cpp23.depr]

Change: Removal of atomic access API for `shared_ptr` objects.

Rationale: The old behavior was brittle. `shared_ptr` objects using the old API were not protected by the type system, and any interaction with code not using this API would silently produce undefined behavior. A complete type-safe replacement is provided in the form of `atomic<shared_ptr<T>>`.

Effect on original feature: Deletion of an old feature where a superior replacement exists within the standard.

Difficulty of converting: Violations will be diagnosed by the C++ translator, as there are no remaining overloads that would match such calls. Violations are addressed by replacing affected `shared_ptr<T>` objects with `atomic<shared_ptr<T>>`.

7.3 Strike wording from Annex D

D.23 [depr.util.smartptr.shared.atomic] Deprecated `shared_ptr` atomic access

¹ The header `<memory>` (20.2.2 [memory.syn]) has the following additions:

```
namespace std {
template <class T>
    bool atomic_is_lock_free(const shared_ptr<T>* p);

template <class T>
    shared_ptr<T> atomic_load(const shared_ptr<T>* p);
template <class T>
    shared_ptr<T> atomic_load_explicit(const shared_ptr<T>* p, memory_order mo);

template <class T>
    void atomic_store(shared_ptr<T>* p, shared_ptr<T> r);
template <class T>
    void atomic_store_explicit(shared_ptr<T>* p, shared_ptr<T> r, memory_order mo);

template <class T>
    shared_ptr<T> atomic_exchange(shared_ptr<T>* p, shared_ptr<T> r);
template <class T>
    shared_ptr<T> atomic_exchange_explicit(shared_ptr<T>* p, shared_ptr<T> r, memory_order mo);

template <class T>
    bool atomic_compare_exchange_weak(
        shared_ptr<T>* p, shared_ptr<T>* v, shared_ptr<T> w);
template <class T>
    bool atomic_compare_exchange_strong(
        shared_ptr<T>* p, shared_ptr<T>* v, shared_ptr<T> w);
template <class T>
    bool atomic_compare_exchange_weak_explicit(
        shared_ptr<T>* p, shared_ptr<T>* v, shared_ptr<T> w,
        memory_order success, memory_order failure);
template <class T>
    bool atomic_compare_exchange_strong_explicit(
        shared_ptr<T>* p, shared_ptr<T>* v, shared_ptr<T> w,
        memory_order success, memory_order failure);
}
```

- 2 Concurrent access to a `shared_ptr` object from multiple threads does not introduce a data race if the access is done exclusively via the functions in this section and the instance is passed as their first argument.
- 3 The meaning of the arguments of type `memory_order` is explained in 33.5.4 [\[atomics.order\]](#).

```
template<class T>
bool atomic_is_lock_free(const shared_ptr<T>* p);
```

- 4 *Preconditions:* `p` shall not be null.
- 5 *Returns:* `true` if atomic access to `*p` is lock-free, `false` otherwise.
- 6 *Throws:* Nothing.

```
template<class T>
shared_ptr<T> atomic_load(const shared_ptr<T>* p);
```

- 7 *Preconditions:* `p` shall not be null.
- 8 *Returns:* `atomic_load_explicit(p, memory_order::seq_cst)`.
- 9 *Throws:* Nothing.

```
template<class T>
shared_ptr<T> atomic_load_explicit(const shared_ptr<T>* p, memory_order mo);
```

- 10 *Preconditions:* `p` shall not be null.
- 11 *Preconditions:* `mo` shall not be `memory_order::release` or `memory_order::acq_rel`.
- 12 *Returns:* `*p`.
- 13 *Throws:* Nothing.

```
template<class T>
void atomic_store(shared_ptr<T>* p, shared_ptr<T> r);
```

- 14 *Preconditions:* `p` shall not be null.
- 15 *Effects:* As if by `atomic_store_explicit(p, r, memory_order::seq_cst)`.
- 16 *Throws:* Nothing.

```
template<class T>
void atomic_store_explicit(shared_ptr<T>* p, shared_ptr<T> r, memory_order mo);
```

- 17 *Preconditions:* `p` shall not be null.
- 18 *Preconditions:* `mo` shall not be `memory_order::acquire` or `memory_order::acq_rel`.
- 19 *Effects:* As if by `p->swap(r)`.
- 20 *Throws:* Nothing.

```
template<class T>
shared_ptr<T> atomic_exchange(shared_ptr<T>* p, shared_ptr<T> r);
```

- 21 *Preconditions:* `p` shall not be null.
- 22 *Returns:* `atomic_exchange_explicit(p, r, memory_order::seq_cst)`.
- 23 *Throws:* Nothing.

```
template<class T>
shared_ptr<T> atomic_exchange_explicit(shared_ptr<T>* p, shared_ptr<T> r, memory_order mo);
```


24 *Preconditions:* `p` shall not be null.

25 *Effects:* As if by `p->swap(r)`.

26 *Returns:* The previous value of `*p`.

27 *Throws:* Nothing.

```
template<class T>
    bool atomic_compare_exchange_weak(shared_ptr<T>* p, shared_ptr<T>* v, shared_ptr<T> w);
```

28 *Preconditions:* `p` shall not be null.

29 *Returns:* `atomic_compare_exchange_weak_explicit(p, v, w, memory_order::seq_cst, memory_order::seq_cst)`.

30 *Throws:* Nothing.

```
template<class T>
    bool atomic_compare_exchange_strong(shared_ptr<T>* p, shared_ptr<T>* v, shared_ptr<T> w);
```

31 *Returns:* `atomic_compare_exchange_strong_explicit(p, v, w, memory_order::seq_cst, memory_order::seq_cst)`.

```
template <class T>
    bool atomic_compare_exchange_weak_explicit(
        shared_ptr<T>* p, shared_ptr<T>* v, shared_ptr<T> w,
        memory_order success, memory_order failure);
template <class T>
    bool atomic_compare_exchange_strong_explicit(
        shared_ptr<T>* p, shared_ptr<T>* v, shared_ptr<T> w,
        memory_order success, memory_order failure);
```

32 *Preconditions:* `p` shall not be null and `v` shall not be null. The `failure` argument shall not be `memory_order::release` nor `memory_order::acq_rel`.

33 *Effects:* If `*p` is equivalent to `*v`, assigns `w` to `*p` and has synchronization semantics corresponding to the value of `success`, otherwise assigns `*p` to `*v` and has synchronization semantics corresponding to the value of `failure`.

34 *Returns:* `true` if `*p` was equivalent to `*v`, `false` otherwise.

35 *Throws:* Nothing.

36 *Remarks:* Two `shared_ptr` objects are equivalent if they store the same pointer value and share ownership. The weak form may fail spuriously. See 33.5.8.2 [\[atomics.types.operations\]](#).

7.4 Update cross-reference for stable labels for C++23

Cross-references from ISO C++ 2023

All clause and subclause labels from ISO C++ 2023 (ISO/IEC 14882:2023, *Programming Language — C++*) are present in this document, with the exceptions described below.

`container.gen.reqmts` *see*

`container.requirements.general`

`depr.res.on.required` *removed*

`depr.util.smartptr.shared.atomic` *removed*

8 Acknowledgements

Thanks to Michael Park for the pandoc-based framework used to transform this document's source from Markdown.

Thanks to Herb Sutter for first bringing this problem to the attention of WG21, along with the proposed solution, a decade ago!

Thanks to Lori Hughes for reviewing this paper and providing editorial feedback.

9 References

[N4958] Thomas Köppe. 2023-08-14. Working Draft, Programming Languages -- C++.
<https://wg21.link/n4958>

[P2139R2] Alisdair Meredith. 2020-07-15. Reviewing Deprecated Facilities of C++20 for C++23.
<https://wg21.link/p2139r2>

[P2863R1] Alisdair Meredith. 2023-08-16. Review Annex D for C++26.
<https://wg21.link/p2863r1>